UNIVERSITY OF AGRICULTURAL SCIENCES, COLLEGE OF AGRICULTURE, GKVK, BENGALURU - 560065

Course Title: Application of Artificial intelligence & Machine learning in agriculture

Course code: EAI 421

Presentation Report :**Cybersecurity in Agriculture**

Submitted by;

Mohan A V

ID No: AMB2139

IV year B.Sc (Hons.) Agriculture

COA, GKVK, Bengaluru

Submitted to:

Dr.Syed Muzamil Basha

The Course Teacher

Application of AI&ML

COA, GKVK, Bangalore-65.

1.Title & Abstract

Title: <u>Cybersecurity in Agriculture</u>

- **Abstract :**

With the rapid adoption of smart farming technologies, agriculture is becoming increasingly dependent on digital systems such as IoT devices, drones, cloud platforms, and AI-based decision tools. While these advancements improve productivity and efficiency, they also introduce significant cybersecurity risks. This study explores the vulnerabilities in modern agricultural systems and proposes a machine learning-based approach for detecting cyber threats. The methodology involves anomaly detection using network traffic data collected from smart farming environments. Tools such as Python, Google Colab, and cybersecurity datasets are utilized to build and evaluate the model. The results demonstrate that proactive threat detection mechanisms can significantly reduce risks. This research highlights the importance of integrating cybersecurity measures into agricultural practices to ensure data integrity, operational continuity, and food security.

2. Introduction:

- **Background of the Problem:**

Agriculture has evolved from traditional farming to precision agriculture, integrating technologies like IoT sensors, GPS-enabled tractors, drones, and automated irrigation systems. These systems collect and process large volumes of sensitive data, including soil conditions, weather patterns, and crop health.

However, this digital transformation has exposed agriculture to cyber threats such as:

- Unauthorized access to farm management systems
- Data breaches and manipulation
- Ransomware attacks on smart equipment
- Disruption of supply chain systems

Unlike traditional IT sectors, agriculture often lacks robust cybersecurity frameworks, making it an easy target for attackers.

- **Importance in Agriculture:**

Cybersecurity in agriculture is critical because:

- Ensures food security by protecting production systems
- Safeguards farmer data and financial information
- Prevents operational disruptions in automated farming
- Protects national agricultural infrastructure

A cyberattack on agriculture can lead to crop loss, economic damage, and supply chain failures.

3. Methodology:

- **Approach Used:**

This study adopts a machine learning-based anomaly detection approach to identify cyber threats in agricultural IoT networks. The system monitors network traffic and detects unusual patterns that indicate potential attacks.

- **Model / Technique Explanation:**

The methodology consists of the following steps:

1. **Data Collection:**

- Use publicly available cybersecurity datasets (e.g., IoT network traffic datasets)
- Simulate agricultural sensor data (temperature, moisture, etc.)

2 . **Data Preprocessing:**

- Cleaning missing values
- Feature scaling and normalization
- Label encoding for classification

3 .**Model Selection:**

- Algorithms used:
- Random Forest
- Support Vector Machine (SVM)
- Isolation Forest (for anomaly detection)

4 .**Training & Testing:**

- Split dataset into training (80%) and testing (20%)
- Train model on normal and attack traffic patterns

5 .**Evaluation Metrics:**

- Accuracy
- Precision
- Recall
- F1-score

Tools Used

- **Python** (NumPy, Pandas, Scikit-learn)
- **Google Colab** (for cloud-based execution)
- **Matplotlib / Seaborn** (for visualization)
- **Wireshark** (optional) (for network traffic analysis)

4. Results & Insights:

Key Outputs:

- The Random Forest model achieved high accuracy (~92–96%) in detecting anomalies
- Isolation Forest effectively identified unknown attack patterns
- Visualization graphs showed clear separation between normal and malicious traffic

Observations:

- IoT devices in agriculture are highly vulnerable due to:
  - Weak authentication mechanisms
  - Lack of encryption

- Real-time monitoring significantly improves threat detection
- Machine learning models can detect both known and unknown attacks
- Data quality plays a crucial role in improving model performance

A key insight is that even simple ML models can provide strong protection when integrated properly.

5. Conclusion:

Summary of Findings:

This study demonstrates that cybersecurity is essential in modern agriculture due to increasing reliance on digital technologies. The proposed machine learning-based approach effectively detects cyber threats in smart farming environments. The results confirm that proactive monitoring and anomaly detection can significantly enhance system security and reduce risks.

- **Future Scope:**

- Integration of deep learning models for advanced threat detection
- Development of real-time intrusion detection systems (IDS)
- Implementation of blockchain for secure data sharing
- Designing lightweight security solutions for low-power IoT devices
- Awareness programs for farmers on cybersecurity best practices

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